```
#include <stdio.h>
#include <stdlib.h>

struct Node
{
    int coeff;
    int exp;
    struct Node *next;
};

struct Node* createHeader()
{
    struct Node *head = (struct Node*)malloc(sizeof(struct Node));
    head->coeff = 0;
    head->exp = -1;
    head->next = head;
    return head;
}

void insertTerm(struct Node *head, int coeff, int exp)
{
    if (coeff == 0)
    {
        return;
    }

    struct Node *newnode = (struct Node*)malloc(sizeof(struct Node));
    newnode->coeff = coeff;
    newnode->exp = exp;

    struct Node *temp = head;
    while (temp->next != head)
    {
        temp = temp->next;
    }

    temp->next = newnode;
    newnode->next = head;
}

void createPolynomial(struct Node *head)
{
    int n, coeff, exp, i;
    printf("Enter the number of terms: ");
    scanf("%d", &n);
```

```
void displayPolynomial(struct Node *head)
{
    if (head == NULL || head->next == head)
    {
        printf("0\n");
        return;
    }

    struct Node *temp = head->next;

    while (temp != head)
    {
        int absCoeff = abs(temp->coeff);

        if (temp->coeff > 0 && temp != head->next)
        {
            printf(" + ");
        }
        else if (temp->coeff < 0 && temp != head->next)
        {
            printf(" - ");
        }
        else if (temp->coeff < 0 && temp == head->next)
        {
            printf("-");
        }

        if (temp->exp == 0)
        {
            printf("%d", absCoeff);
        }
        else if (temp->exp == 1)
        {
            if (absCoeff == 1)
            {
                printf("x");
            }
            else
            {
                printf("%dx", absCoeff);
            }
        }
        else
        {
            if (absCoeff == 1)
```

```
struct Node* addPolynomial(struct Node *h1, struct Node *h2)
{
    struct Node *h3 = createHeader();
    struct Node *t1 = h1->next;
    struct Node *t2 = h2->next;

    while (t1->exp != -1 || t2->exp != -1)
    {
        if (t1->exp == t2->exp)
        {
            int sum = t1->coeff + t2->coeff;
            if (sum != 0)
            {
                insertTerm(h3, sum, t1->exp);
            }
            t1 = t1->next;
            t2 = t2->next;
        }
        else if (t1->exp > t2->exp)
        {
            insertTerm(h3, t1->coeff, t1->exp);
            t1 = t1->next;
        }
        else
        {
            insertTerm(h3, t2->coeff, t2->exp);
            t2 = t2->next;
        }
    }
    return h3;
}

int main()
{
    struct Node *poly1 = createHeader();
    struct Node *poly2 = createHeader();
    struct Node *poly3 = NULL;
    int choice;

    while (1)
    {
        printf("\n--- POLYNOMIAL OPERATIONS (CLL) ---\n");
        printf("1. Create Polynomial 1\n");
        printf("2. Create Polynomial 2\n");
        printf("3. Display Polynomial 1\n");
```

```
        switch (choice)
        {
            case 1:
                printf("\nCreate Polynomial 1:\n");
                createPolynomial(poly1);
                break;

            case 2:
                printf("\nCreate Polynomial 2:\n");
                createPolynomial(poly2);
                break;

            case 3:
                printf("\nPolynomial 1: ");
                displayPolynomial(poly1);
                break;

            case 4:
                printf("\nPolynomial 2: ");
                displayPolynomial(poly2);
                break;

            case 5:
                poly3 = addPolynomial(poly1, poly2);
                printf("\nResultant Polynomial (Sum): ");
                displayPolynomial(poly3);
                break;

            case 6:
                exit(0);

            default:
                printf("\nInvalid choice! Please try again.\n");
        }
    }

    return 0;
}
```